Math 1232 Summer 2025
Practice Final

- You will have 90 minutes for this test.
- You are not allowed to consult books or notes during the test, but you may use a one-page, **two-sided,** handwritten cheat sheet you've made for yourself ahead of time.
- You may not use a calculator. You may leave answers unsimplified, except you should compute trigonometric functions as far as possible.
- The exam has 12 problems, one on each mastery topic we've covered. You must do all of the problems on major topics, but need only complete 4 to 6 of the problems on secondary topics—your best 4 will count toward the exam grade, and up to an additional 2 problems can get you 1 or 2 bonus points.
- Each part of each topic is worth ten points. The whole test is scored out of 120 points.
- Read the questions carefully and make sure to answer the actual question asked. Make sure to justify your answers—math is largely about clear communication and argument, so an unjustified answer is much like no answer at all. When in doubt, show more work and write complete sentences.
- If you need more paper to show work, I have extra at the front of the room.
- Good luck!

	a	b		a	b
M1:			M2:		
M3:			M4		
S1:		S2:		S3	
S4:		S5:		S6	
S7:		S8:		Total:	

Name: Solutions

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Problem 1 (M1).

(a) Compute $\frac{d}{dt} 2^t \log_5(t)$.

Solution.

$$\frac{d}{dt} 2^t \log_5(t) = 2^t \ln(2) \log_5(t) + 2^t \frac{1}{t \ln(5)}$$

from the product rule.

(b) Compute $\int_0^{2^{-\frac{1}{3}}} \frac{x^2}{\sqrt{1-x^6}}\, dx$.

Solution. The obvious substitution is $u = 1 - x^6$ but that doesn't really work. After experimenting / noticing we really need something to cancel the x^2 in the numerator, we see $u = x^3$ gives $du = 3x^2\, dx$ and thus

$$\begin{aligned}
\int_0^{2^{-\frac{1}{3}}} \frac{x^2}{\sqrt{1-x^6}}\, dx &= \int_0^{1/2} \frac{1}{3} \frac{1}{\sqrt{1-u^2}}\, du \\
&= \frac{1}{3} \arcsin(u)\Big|_0^{1/2} \\
&= \frac{1}{3} (\arcsin(1/2) - \arcsin(0)) = \frac{1}{3} (\pi/6 - 0) = \frac{\pi}{18}.
\end{aligned}$$

Alternatively we could compute that

$$\int \frac{x^2}{\sqrt{1-x^6}}\, dx = \frac{1}{3} \arcsin(x^3) + C$$

and thus

$$\begin{aligned}
\int_0^{2^{-\frac{1}{3}}} \frac{x^2}{\sqrt{1-x^6}}\, dx &= \frac{1}{3} \arcsin(x^3)\Big|_0^{2^{-\frac{1}{3}}} \\
&= \frac{1}{3} (\arcsin(1/2) - \arcsin(0)) = \frac{1}{3} (\pi/6 - 0) = \frac{\pi}{18}.
\end{aligned}$$

Name: Solutions

Problem 2 (M2). Compute the following integrals using any of the methods we have learned in class. Show enough work to justify your answers.

(a) $\displaystyle\int \frac{x^3}{\sqrt{x^2+4}}\,dx$

Solution. Here we should do a trigonometric substitution. We take $x = 2\tan(\theta)$, so we have $dx = 2\sec^2(\theta)\,d\theta$. Then

$$\begin{aligned}
\int \frac{x^3}{\sqrt{x^2+4}}\,dx &= \int \frac{8\tan^3(\theta)}{\sqrt{4\tan^2(\theta)+4}} 2\sec^2(\theta)\,d\theta = \int \frac{8\tan^3(\theta)}{\sqrt{4\sec^2(\theta)}} 2\sec^2(\theta)\,d\theta \\
&= \int \frac{8\tan^3(\theta)}{2\sec(\theta)} 2\sec^2(\theta)\,d\theta = \int 8\sec(\theta)\tan^3(\theta)\,d\theta \\
&= \int 8\sec(\theta)\tan(\theta)(\sec^2(\theta)-1)\,d\theta; \qquad u = \sec(\theta),\ du = \sec(\theta)\tan(\theta)\,d\theta \\
&= 8\int u^2 - 1\,du = \frac{8}{3}u^3 - 8u + C = \frac{8}{3}\sec^3(\theta) - 8\sec(\theta) + C.
\end{aligned}$$

Now we just need to figure out what $\sec(\theta)$ is. We know $\tan(\theta) = x/2$, so we get a triangle with opposite side x, adjacent side 2, and hypotenuse $\sqrt{x^2+4}$. Then we see that $\sec(\theta) = \frac{\sqrt{x^2+4}}{2}$, and thus the integral is

$$\int \frac{x^3}{\sqrt{x^2+4}}\,dx = \frac{(x^2+4)^{3/2}}{3} - 4\sqrt{x^2+4} + C.$$

(b) $\displaystyle\int x^2 e^{3x}\,dx$

Solution. We use integration by parts. Take $u = x^2, dv = e^{3x}\,dx$ so $du = 2x\,dx, v = e^{3x}/3$. Then

$$\begin{aligned}
\int x^2 e^{3x}\,dx &= x^2\frac{e^{3x}}{3} - \int \frac{2}{3}xe^{3x}\,dx \\
&= \frac{x^2e^{3x}}{3} - \frac{2}{3}\left(\frac{1}{3}xe^{3x} - \int \frac{1}{3}e^{3x}\right) \\
&= \frac{x^2e^{3x}}{3} - \frac{2xe^{3x}}{9} + \frac{2e^{3x}}{27} + C.
\end{aligned}$$

where we have done integration by parts a second time with $u = x$ and $dv = e^{3x}$.

Name: Solutions

Problem 3 (M3). Remember to show *all* your work—this includes stating any convergence tests you use, as well as computing any limits that arise.

(a) Analyze the convergence of the series $\displaystyle\sum_{n=1}^{\infty} \frac{(-2)^n}{n^3+n}$.

Solution. We use the Ratio test. We have

$$\begin{aligned}\lim_{n\to\infty} \left| \frac{\frac{(-2)^{n+1}}{(n+1)^3+n+1}}{\frac{(-2)^n}{n^3+n}} \right| &= \lim_{n\to\infty} \frac{2(n^3+n)}{(n+1)^3+n+1} \\ &= \lim_{n\to\infty} \frac{2n^3+\text{stuff}}{n^3+\text{stuff}} = 2.\end{aligned}$$

This limit (2) is greater than 1, so by the ratio test this diverges.

Alternatively, we could note that $\lim_{n\to\infty} \frac{(-2)^n}{n^3+n} = \pm\infty$, so by the divergence test this diverges. But it's a little tricky to cleanly argue that this goes to infinity; we can't really use L'Hospital's rule without getting the negative sign out of there somehow. Still, I would accept this as an answer!

(b) Analyze the convergence of the series $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n}{5n^2+3n}$.

Solution. This clearly converges by the alternating series test, since $\lim_{n\to\infty} \frac{1}{5n^2+3n} = 0$, but does it absolutely converge? The ratio test won't work; if we work it out we'll get a limit of 1. But we have

$$\sum_{n=1}^{\infty} \left| \frac{(-1)^n}{5n^2+3n} \right| = \sum_{n=1}^{\infty} \frac{1}{5n^2+3n},$$

so we can use the Limit Comparison Test with $b_n = \frac{1}{n^2}$. We compute

$$\lim_{n\to\infty} \frac{\frac{1}{5n^2+3n}}{\frac{1}{n^2}} = \lim_{n\to\infty} \frac{n^2}{5n^2+3n} = 5.$$

This is a nonzero real number, so since $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges by the p-test ($p = 2$), by the Limit Comparison Test, $\sum_{n=1}^{\infty} \frac{1}{5n^2+3n}$ converges. Thus our original series converges absolutely. (And thus we didn't actually need to check for whether the alternating series test applies.)

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Problem 4 (M4).

(a) Let $f(x) = \cos^2(x)$. Use *the definition of a Taylor series* to find $T_4(x, \pi)$ for this function. (That is, find the terms up through the degree four term of the series centered at $a = \pi$.)

Solution.

$$\begin{aligned} f(x) &= \cos^2(x) & f(\pi) &= 1 \\ f'(x) &= -2\cos(x)\sin(x) & f'(\pi) &= 0 \\ f''(x) &= 2\sin^2(x) - 2\cos^2(x) & f''(\pi) &= -2 \\ f'''(x) &= 4\sin(x)\cos(x) + 4\cos(x)\sin(x) & f'''(\pi) &= 0 \\ f^{(4)} &= 8\cos^2(x) - 8\sin^2(x) & f^{(4)}(\pi) &= 8. \end{aligned}$$

So we have

$$T_4(x, \pi) = 1 - \frac{2}{2!}(x-\pi)^2 + \frac{8}{4!}(x-\pi)^4 = 1 - (x-\pi)^2 + \frac{1}{3}(x-\pi)^4.$$

(b) Write a power series expression for $\frac{x^4}{2-4x}$ centered a 0. What is the radius of convergence?

Solution. We know that

$$\begin{aligned} \frac{1}{1-2x} &= \sum_{n=0}^{\infty}(2x)^n \\ \frac{x^4}{2} \cdot \frac{1}{1-2x} &= \frac{x^4}{2}\sum_{n=0}^{\infty}(2x)^n = \sum_{n=0}^{\infty}\frac{2^n}{2}x^{n+4} \\ \text{(or)} \quad &= \sum_{n=4}^{\infty} 2^{n+3}x^n. \end{aligned}$$

The radius of convergence is $1/2$. We can figure that out by reasoning from the geometric series: the radius of convergence for the geometric series is 1, so it converges for $-1 < 2x < 1$ or $-1/2 < x < 1/2$. Or we can use the ratio test:

$$\lim_{n\to\infty}\left|\frac{2^{n+4}x^{n+1}}{2^{n+3}x^n}\right| = \lim_{n\to\infty}|2x|$$

and thus it converges when $2|x| < 1$.

Name: Solutions

Problem 5 (S1). Let $f(x) = \sqrt[4]{x^7 + 5x^5 + 4x^3 + 6}$. Find $(f^{-1})'(2)$.

Solution. Plugging in numbers, we see that $f(0) = \sqrt[4]{6}$ but $f(1) = \sqrt[4]{16} = 2$, so $f^{-1}(2) = 1$. We compute

$$\begin{aligned} f'(x) &= \frac{1}{4}(x^7 + 5x^5 + 4x^3 + 6)^{-3/4}(7x^6 + 25x^4 + 12x^2) \\ f'(1) &= \frac{1}{4}(16)^{-3/4}(7 + 25 + 12) = \frac{44}{32} = \frac{11}{8} \end{aligned}$$

and thus by the Inverse Function Theorem,

$$(f^{-1})'(2) = \frac{1}{f'(f^{-1}(2))} = \frac{1}{f'(1)} = \frac{1}{11/8} = \frac{8}{11}.$$

Problem 6 (S2). Compute $\lim_{x\to 0} \frac{\arctan(x)}{\ln(x+1)}$.

Solution. We see the top and bottom both approach zero, so we use L'Hôpital's Rule:

$$\lim_{x\to 0} \frac{\arctan(x)}{\ln(x+1)} \overset{\text{L'H}}{=} \lim_{x\to 0} \frac{1/(x^2+1)}{1/(x+1)} = \lim_{x\to 0} \frac{x+1}{x^2+1} = 1.$$

Name: Solutions

Problem 7 (S3). Compute $\int_1^5 \frac{1}{\sqrt{x-1}}\,dx$.

Solution.

$$\begin{aligned}\int_1^5 \frac{1}{\sqrt{x-1}}\,dx &= \lim_{t\to 1^+}\int_t^5 \frac{1}{\sqrt{x-1}}\,dx\\ &= \lim_{t\to 1^+} 2\sqrt{x-1}\,\big|_t^5\\ &= \lim_{t\to 1^+} 2\sqrt{4}-2\sqrt{t-1} = 4.\end{aligned}$$

(Using the formalism of the limit here is important—this is an improper integral due to the discontinuity at $x = 1$, and we must treat it as such!)

Problem 8 (S4). Compute the area of the surface obtained by taking the curve $x^{2/3}+y^{2/3} = 1$ as x goes from 0 to 1 and rotating it around the x-axis.

Solution. We have $y = (1-x^{2/3})^{3/2}$, so

$$\begin{aligned}y' &= \frac{3}{2}(1-x^{2/3})^{1/2}\cdot\frac{-2}{3}x^{-1/3}\\ &= \sqrt{1-x^{2/3}}x^{-1/3}\\ (y')^2 &= (1-x^{2/3})x^{-2/3} = x^{-2/3}-1\\ \sqrt{1+(y')^2} &= \sqrt{x^{-2/3}} = x^{-1/3}.\end{aligned}$$

Then we compute

$$\begin{aligned}L &= \int_0^1 2\pi(1-x^{\frac{2}{3}})^{\frac{3}{2}}\sqrt{1+(y')^2}\,dx\\ &= 2\pi\int_0^1 (1-x^{\frac{2}{3}})^{\frac{3}{2}}\cdot x^{-1/3}\,dx\end{aligned}$$

At this point we have the substitution $u = 1-x^{\frac{2}{3}}$ with $du = -\frac{2}{3}x^{\frac{-1}{3}}\,dx$ to get

$$\begin{aligned}&= 2\pi\int_1^0 -\frac{3}{2}u^{\frac{3}{2}}\,du = 3\pi\int_0^1 u^{\frac{3}{2}}\,du = 3\pi\frac{2}{5}u^{\frac{5}{2}}\Big|_0^1\\ &= \frac{6\pi}{5}\left(1^{5/2}-0^{5/2}\right) = \frac{6\pi}{5}.\end{aligned}$$

Name: Solutions

Problem 9 (S5). Find the (particular) solution to the initial value problem given by $-2x + 4y^3\sqrt{x^2+3} \cdot y' = 0$ and $y(1) = 1$.

Solution.

$$\begin{aligned} 4y^3 y' \sqrt{x^2+3} &= 2x \\ 4y^3\, dy &= \frac{2x}{\sqrt{x^2+3}}\, dx \\ y^4 &= 2\sqrt{x^2+3} + C \\ y &= \sqrt[4]{2\sqrt{x^2+3} + C}. \end{aligned}$$

Then we have

$$\begin{aligned} 1 &= \sqrt[4]{2\sqrt{4} + C} = \sqrt[4]{4+C} \\ 1 &= 4 + C \\ C &= -3 \\ y &= \sqrt[4]{2\sqrt{x^2+3} - 3}. \end{aligned}$$

Problem 10 (S6). Compute $\displaystyle\sum_{n=1}^{\infty} \ln\left(\frac{n+4}{n+3}\right)$.

Solution. This is the same as

$$\begin{aligned} \sum_{n=1}^{\infty} \ln(n+4) - \ln(n+3) &= \lim_{k\to\infty} \sum_{n=1}^{k} \ln(n+4) - \ln(n+3) \\ &= \lim_{k\to\infty} \big(\ln(5) - \ln(4)\big) + \big(\ln(6) - \ln(5)\big) + \cdots + \big(\ln(k+4) - \ln(k+3)\big) \\ &= \lim_{k\to\infty} \ln(k+4) - \ln(4) = \infty. \end{aligned}$$

Thus this sum diverges to infinity.

Name: Solutions

Problem 11 (S7). Find the radius and interval of convergence of the series $\sum_{n=1}^{\infty} \frac{(5x-3)^n}{\sqrt{n}}$.

Solution. We use the ratio test.

$$\lim_{n\to\infty} \left| \frac{\frac{(5x-3)^{n+1}}{\sqrt{n+1}}}{\frac{(5x-3)^n}{\sqrt{n}}} \right| = \lim_{n\to\infty} \left| \frac{(5x-3)\sqrt{n}}{\sqrt{n+1}} \right|$$
$$= |5x-3| \lim_{n\to\infty} \frac{\sqrt{n}}{\sqrt{n+1}} = |5x-3|.$$

So we need $|5x-3| < 1$ or $-1 < 5x-3 < 1$, or $2 < 5x < 4$ or $2/5 < x < 4/5$. We need to have x in the interval $(3/5 - 1/5, 3/5 + 1/5)$, so the radius is $1/5$.

To find the interval we need to check the endpoints. We see

$$\sum_{n=0}^{\infty} \frac{(2-3)^n}{\sqrt{n}} \sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n}}$$

converges by alternating series test

$$\sum_{n=0}^{\infty} \frac{(4-3)^n}{\sqrt{n}} = \sum_{n=0}^{\infty} \frac{1}{\sqrt{n}}$$

diverges by p-series test

Thus the interval of convergence is $[2/5, 4/5)$.

Problem 12 (S8). Find a (non-parametric) equation of the line tangent to the curve parametrized by $x = \cos^3(t), y = \sin^3(t)$ at the point $(1/8, -3\sqrt{3}/8)$.

Solution. We have $x'(t) = -3\cos^2(t)\sin(t)$ and $y'(t) = 3\sin^2(t)\cos(t)$. This point happens at $t = -\pi/3$, so we have

$$x'(-\pi/3) = 3\sqrt{3}/8$$
$$y'(-\pi/3) = 9/8$$
$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{9/8}{3\sqrt{3}/8} = \sqrt{3}$$
$$y + \frac{3\sqrt{3}}{8} = \sqrt{3}(x - 1/8).$$